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Build and Deploy Machine Learning Solutions on Vertex AI

Course · 7 Complete hours

Course overview

Build and Deploy Machine Learning Solutions on Vertex AI

Identify Damaged Car Parts with Vertex AutoML Vision

Deploy a BigQuery ML Customer Churn Classifier to Vertex AI for Online Predictions

Vertex AI Pipelines: Qwik Start

Build and Deploy Machine Learning Solutions on Vertex AI: Challenge Lab

Your Next Steps

Lab setup instructions and requirements

Protect your account and progress. Always use a private browser window and lab credentials to run this lab.

End Lab 01:29:42

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username student-82-34585b254959f

Password df1caN1CMSZc

Project ID qwiklabs-gcp-08-31772f4f

Region us-central1

Zone us-central1-c

Deploy a BigQuery ML Customer Churn Classifier to Vertex AI for Online Predictions

Lab 1 hour 30 minutes No cost Intermediate

★★★★☆ Rate Lab

This lab may incorporate AI tools to support your learning.

GSP944

Google Cloud Self-Paced Labs

Overview

Lab instructions and tasks 0/100

GSP944

Overview

Objectives

Setup and requirements

Task 1. Create a Vertex AI Workbench instance

Task 2. Copy the notebook from a Cloud Storage bucket

Task 3. Create a BigQuery dataset

Task 4. Create a BigQuery ML XGBoost churn propensity model

Task 5. Evaluate your BigQuery ML model

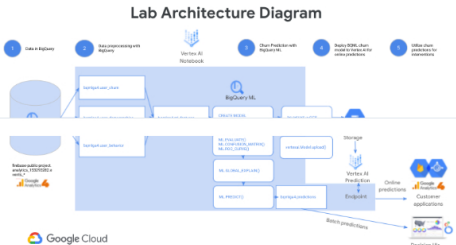
Task 6. Batch predict user churn with your BigQuery ML model

Congratulations!

In this lab, you will train, tune, evaluate, explain, and generate batch and online predictions with a BigQuery ML XGBoost model. You will use a Google Analytics 4 dataset from a real mobile application, Flood it! ([Android app](#), [iOS app](#)), to determine the likelihood of users returning to the application. You will generate batch predictions with your BigQuery ML model as well as export and deploy it to **Vertex AI** for online predictions using the Vertex AI Python SDK.

[BigQuery ML](#) lets you train and do batch inference with machine learning models in BigQuery using standard SQL queries faster by eliminating the need to move data with fewer lines of code.

[Vertex AI](#) is Google Cloud's complimentary next generation, unified platform for machine learning development. By developing and deploying BigQuery ML machine learning solutions on Vertex AI, you can leverage a scalable online prediction service and MLOps tools for model retraining and monitoring to significantly enhance your development productivity, the ability to scale your workflow and decision making with your data, and accelerate time to value.



Note: BQML is now BigQuery ML.

This lab is inspired by and extends Churn prediction for game developers using [Google Analytics 4 \(GA4\)](#) and [BigQuery ML](#). Read the blog post and accompanying tutorial for additional depth on this use case and BigQuery ML.

In this lab, you will go one step further and focus on how Vertex AI extends BigQuery ML's capabilities through online prediction so you can incorporate both customer churn predictions into decision making UIs such as [Looker dashboards](#) but also online predictions directly into customer applications to power targeted interventions such as targeted incentives.

Objectives

In this lab, you learn how to:

- retraining.
- Train a [BigQuery ML XGBoost](#) classifier to predict user churn on a mobile gaming application.
- Tune a BigQuery ML XGBoost classifier using [BigQuery ML hyperparameter tuning features](#).
- Evaluate the performance of a BigQuery ML XGBoost classifier.
- Explain your XGBoost model with [BigQuery ML Explainable AI](#) global feature attributions.
- Generate batch predictions with your BigQuery ML XGBoost model.
- Export a BigQuery ML XGBoost model to a [Google Cloud Storage](#) bucket.
- Upload and deploy a BigQuery ML XGBoost model to a [Vertex AI Prediction Endpoint](#) for online predictions.

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a

you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud account, you may incur charges to that account.

Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

2. Click through the following windows:

- Continue through the Cloud Shell information window.
- Authorize Cloud Shell to use your credentials to make Google Cloud API calls.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `qwiklabs-gcp-00-31772f4cd8fe`. The output contains a line that declares the **Project_ID** for this session:

```
Your Cloud Platform project in this session is set to quiklabs-gcp-00-31772f4cd8fe
```

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

3. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

4. Click **Authorize**.

```
ACTIVE: *
ACCOUNT: student-02-34585b254959@quiklabs.net

To set the active account, run:
$ gcloud config set account "ACCOUNT"
```

5. (Optional) You can list the project ID with this command:

```
gcloud config list project
```

Output:

```
[core]
project = quiklabs-gcp-00-31772f4cd8fe
```

Note: For full documentation of `gcloud`, in Google Cloud, refer to [the gcloud CLI overview guide](#).

Task 1. Create a Vertex AI Workbench instance

1. In the Google Cloud console, from the **Navigation menu** () , select **Vertex AI > Dashboard**.
2. Click **Enable All Recommended APIs**.
3. On the left-hand side, click **Workbench**.
4. At the top of the Workbench page, ensure you are in the **Instances** view.
5. Click  **Create New**.
6. **Configure the Instance:**
 - **Name:** lab-workbench
 - **Region:** Set the region to `us-central1`
 - **Zone:** Set the zone to `us-central1-c`
 - **Advanced Options** (Optional): If needed, click "Advanced Options" for further customization (e.g., machine type, disk size)

← Create instance

Details

Environment

Machine type

Disks

Networking

IAM and security

System health

Name *

lab-workbench

Must start with a letter followed by up to 47 lowercase letters, numbers, or hyphens (-) and cannot end with a hyphen

Region *

us-east14 (Northern Virginia)

Zone *

us-east14-b

Some regions are restricted due to a policy set by your organization. [Learn more](#)

☒ Enable Dataproc Serverless Interactive Sessions

Enable access to Dataproc Spark kernels

Labels

+ ADD LABEL

7. Click **Create**.

Note: The instance will take a few minutes to create. A green checkmark will appear next to its name when it's ready.

8. Click **Open JupyterLab** next to the instance name to launch the JupyterLab interface. This will open a new tab in your browser.

Workbench

CREATE NEW

REFRESH

INSTANCESEXECUTIONSCHEDULES

Filter

Instance name	Zone	Auto upgrade	Version	Machine Type	GPUs
lab-workbench	us-east14-b	—	MT24	Efficient Instance 4 vCPU, 16 GB RAM	None

9. Click the **Terminal** icon to open a terminal window.

Launcher

Filter files by name

name

notebook_template...

Untitled.ipynb

Notebook

Python 3 (system)

Python 3.10 (base)

TensorFlow 2.11 (Linux)

Console

Python 3 (system)

Python 3.10 (base)

TensorFlow 2.11 (Linux)

Other

Terminal

Text File

Markdown File

Python File

Show Content Help

to interact with your Workbench instance.

Terminal 1

Filter files by name

name

notebook_template...

Untitled.ipynb

Terminal

Text File

Markdown File

Python File

Show Content Help

Click **Check my progress** to verify the objective.

Create a Vertex AI Workbench Instance

Check my progress

Task 2. Copy the notebook from a Cloud Storage bucket

gcloud storage cp gs://qwiklabs-gcp-00-31772f4cd8fe-labconfig-buck

Click **Check my progress** to verify the objective.

Copy the notebook from a Cloud Storage bucket

Check my progress

- Open the notebook file **lab_exercise.ipynb**.
- In the **Select Kernel** dialog, choose **TensorFlow 2-11** from the list of available kernels.
- In the **Define constants** section, update your **REGION** variable with the **us-central1**.
- Continue the lab in the notebook, and run each cell by clicking the **Run** (▶) icon at the top of the screen. Alternatively, you can execute the code in a cell with **SHIFT + ENTER**.

Read the narrative and make sure you understand what's happening in each cell. As you

graded exercises.

Task 3. Create a BigQuery dataset

Click **Check my progress** to verify the objective.



Create a BigQuery dataset

Check my progress

Task 4. Create a BigQuery ML XGBoost churn propensity model

Click **Check my progress** to verify the objective.



Create a BigQuery ML XGBoost churn propensity model

Check my progress

Task 5. Evaluate your BigQuery ML model

Click **Check my progress** to verify the objective.



Evaluate your BigQuery ML model

Check my progress

Task 6. Batch predict user churn with your BigQuery ML model

Click **Check my progress** to verify the objective.



Batch predict user churn with your BigQuery ML model

Check my progress

Congratulations!

In this lab you trained, tuned, explained, and deployed a BigQuery ML user churn model to Vertex AI to generate high business impact batch and online churn predictions to target customers likely to churn with interventions such as in-game rewards and reminder notifications.

Next steps / Learn more

Read more about Vertex AI in the [Vertex AI Documentation](#).

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. [Our classes](#) include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. [Certifications](#) help you validate and prove your skill and expertise in Google Cloud technologies.

Manual Last Updated July 11, 2025

Lab Last Tested July 11, 2025

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